Security Risk Assessment and Hardening Recommendations.

Prepared by: Crystal Mitchell
Portfolio Version: July 2026

Scenario

This project is based on a cybersecurity training scenario completed during the Google Cybersecurity Professional Certificate program. The objective was to assess an organization's security posture, identify key security risks, and recommend practical hardening controls to reduce risk and improve overall security. This portfolio version has been refined to demonstrate professional security documentation and analytical thinking.

Executive Summary

This assessment identifies key security risks within the environment and recommends controls to strengthen authentication, protect network communications, and improve threat detection capabilities.

Security Risks Identified

- Identified Risks
- Weak authentication practices
- Insecure network communications
- Limited network threat visibility

Security Recommendations

1. Multi-Factor Authentication (MFA)

Risk Addressed:
Credential theft and unauthorized account access.

Recommendation:
Require MFA for all users and enforce strong password policies.

Expected Benefit:
Reduces the likelihood of account compromise even if passwords are stolen.

2. Secure Network Protocols

Risk Addressed:
Sensitive information transmitted over insecure channels.

Recommendation:
Implement HTTPS, SSH, and SFTP while disabling insecure legacy protocols.

Expected Benefit:
Protects confidentiality and integrity of data during transmission.

3. Intrusion Detection System (IDS)

Risk Addressed:
Delayed detection of malicious activity.

Recommendation:
Deploying an IDS to monitor network traffic and alert administrators to suspicious behavior.

Expected Benefit:
Improves visibility into threats and supports faster incident response.

Conclusion

Implementing these recommendations will strengthen authentication, protect sensitive communications, and improve the organization's ability to detect malicious activity. Together these controls reduce risk and establish a stronger foundation for a layered security strategy.

Skills Demonstrated

- Security Risk Assessment
- Security Hardening
- Multi-Factor Authentication (MFA)
- Network Security
- Intrusion Detection Systems (IDS)
- Risk Analysis
- Security Documentation